

Ridge

Nguyen Phu Nhan

Prerequisites: Linear regression.

We define and process $\mathbf{X}$, $\mathbf{y}$, $\mathbf{w}$ similar to linear regression.

Ridge regression extends linear regression by adding an L_2 regularization term to the loss function. The resulting optimization problem is

$$\hat{\mathbf{w}} = \underset{\mathbf{w}}{\operatorname{argmin}} \left(\|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda \|\mathbf{w}\|_2^2 \right),$$

where $\lambda \geq 0$ controls the strength of regularization.

To find the optimal coefficient vector, we differentiate the objective function with respect to $\mathbf{w}$ and set the gradient equal to zero:

$$-2\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{w}) + 2\lambda\mathbf{w} = 0.$$

Rearranging the terms gives

$$(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})\mathbf{w} = \mathbf{X}^\top\mathbf{y}.$$

Therefore, the ridge regression solution is

$$\hat{\mathbf{w}} = (\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^\top\mathbf{y}.$$